

Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry

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Abstract

Multi-agent systems are increasingly used for forecasting future events, on the assumption that deliberation among multiple LLMs improves reasoning and calibration. Existing approaches, however, overlook a basic design choice: *what information each agent receives*. When all agents observe identical evidence, deliberation collapses into herding rather than genuine belief revision, and the ensemble offers little over a single agent. We propose *designed information asymmetry*: partitioning the evidence into a shared public pool and disjoint private subsets, so that each agent holds exclusive knowledge that can reach the others only through deliberation. We show theoretically that this partition reduces inter-agent error correlation, and instantiate it in **InfoDelphi**, a framework combining relevance-aware evidence routing, rationale-based iterative deliberation, and a calibrated aggregation layer that counteracts systematic LLM overconfidence via asymmetric shrinkage. On POLYGYM, a controlled benchmark of 250 resolved Polymarket questions with fixed pre-retrieved evidence, and on 231 questions from FutureX, InfoDelphi attains Brier scores of 0.158 and 0.201, outperforming nine single-agent and multi-agent baselines under a paired bootstrap test on both datasets, with a 13–18% relative Brier improvement over the strongest baseline. Analysis shows that deliberation under homogeneous input collapses forecast diversity while partitioned evidence preserves it, and that overconfidence is best corrected in the aggregation layer: prompt-level skepticism depresses the entire forecast distribution and hurts accuracy.

1 Introduction

Prediction markets such as Polymarket¹ aggregate the beliefs of thousands of participants into real-time probability estimates for future events,

¹<https://polymarket.com/>

achieving forecasting accuracy that often rivals or exceeds expert panels (Tetlock, 2005). With every question resolved against a definitive ground truth, these markets provide a natural benchmark for machine intelligence: *can an automated system reason over evidence and match human-level forecasting?* Recent work has shown that large language models (LLMs) equipped with retrieved evidence can approach human-level performance on binary forecasting tasks (Halawi et al., 2024; Turtel et al., 2025), but these advances rely primarily on single-agent, single-pass pipelines that treat the entire evidence corpus as a monolithic context window. A natural extension is the *multi-agent* setting, where a panel of LLMs deliberates over the evidence before producing a collective forecast. Such frameworks draw inspiration from well-studied human mechanisms for aggregating distributed knowledge—expert committees, structured interviews, and the Delphi method (Dalkey, 1969)—and have shown promise in general reasoning tasks (Chen et al., 2024; Wang et al., 2024).

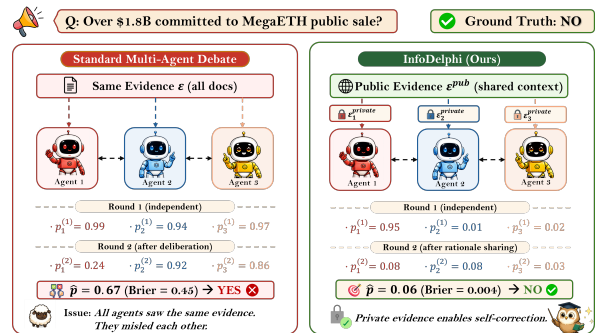


Figure 1: Effect of information asymmetry on a real prediction. **Left:** agents with identical evidence converge to the same incorrect forecast. **Right:** agents with complementary private evidence self-correct through rationale sharing.

Effective collective reasoning, however, requires that participants hold *informationally diverse* views. As formalized in the Condorcet

jury tradition and established empirically for human forecasters (Lorenz et al., 2011), when agents share identical information, deliberation fails to improve collective accuracy: they reach the same conclusions independently, and iterative exchange merely reinforces the shared prior. In LLM systems this problem runs deeper. Recent work shows that models from different providers make correlated errors at rates above 60% (Kim et al., 2025), and that standard multi-agent debate under homogeneous input behaves as a *martingale* whose expected accuracy does not improve over rounds (Choi et al., 2026a). Figure 1 illustrates this failure mode on a real prediction market question: when all agents receive identical evidence, they converge to the same incorrect forecast (left), whereas agents equipped with complementary private evidence can self-correct through deliberation (right).

To address this, we propose *designed information asymmetry*, treating information asymmetry not as a limitation to overcome but as a principle to exploit. We introduce **InfoDelphi**, a multi-agent forecasting framework that partitions evidence into a shared public pool and disjoint private subsets, giving each agent distinct expertise while preserving common ground for communication. This mirrors effective human expert panels, where participants contribute complementary knowledge rather than reading the same briefing materials. InfoDelphi constructs informative evidence partitions using BM25 relevance ranking; during deliberation, agents exchange rationale excerpts rather than raw evidence, enabling efficient propagation of private signals. Beyond information structure, we find that LLM forecasters are systematically overconfident on the affirmative side of low-base-rate questions, and that this bias is best corrected *after* deliberation: the final aggregation layer applies an asymmetric shrinkage toward a NO-leaning prior, with parameters validated across datasets.

We evaluate on POLYGYM, a controlled benchmark of 250 resolved binary Polymarket questions with fixed, pre-retrieved evidence shared across all methods, and on 231 resolved questions from the FutureX benchmark (Zeng et al., 2025). Against nine single-agent and multi-agent baselines run with the same backbone model and the same evidence, InfoDelphi achieves the best Brier score on both datasets (0.158 and 0.201), and a paired bootstrap over questions confirms that every pair-

wise gap is significant at the 95% level. Our analysis further traces *why*: deliberation under homogeneous input collapses forecast diversity (herding), prompt-level skepticism cannot target overconfidence where it occurs, and for an evidence-limited domain (sports), even targeted retrieval augmentation yields no measurable gain.

Our main contributions are as follows:

- We propose *designed information asymmetry* for multi-agent forecasting, and theoretically show that public/private evidence partitioning reduces inter-agent error correlation.
- We introduce **InfoDelphi**, which instantiates this principle through relevance-aware evidence routing, rationale-based deliberation, and a calibrated shrinkage aggregation layer whose parameters transfer across datasets.
- On POLYGYM-250 and FutureX-231, InfoDelphi outperforms nine baselines with paired-bootstrap significance; ablations attribute the gains to information asymmetry and the aggregation layer, and deliberation-dynamics analysis provides direct empirical evidence of herding under homogeneous input.

2 Related Work

LLM-based Forecasting. Early work established a substantial gap between LLM and human forecasting accuracy. Zou et al. (2022) find that language models achieve only 65% accuracy on Autocast compared to 92% for human aggregates, while Schoenegger and Park (2023) show that GPT-4 without augmentation does not significantly outperform a random baseline on Metaculus. Retrieval-augmented generation (Lewis et al., 2020) substantially closes this gap: Yan et al. (2023) improve Autocast accuracy through zero-shot re-ranking and summarization of retrieved news, and Halawi et al. (2024) present a full pipeline achieving near-human performance on Polymarket and Metaculus. More recent work focuses on training: Turtel et al. (2025) fine-tune on 12,100 resolved Polymarket questions, and Chandak et al. (2025) train an 8B model via reinforcement learning that matches proprietary LLMs on prediction benchmarks. On the aggregation side, Schoenegger et al. (2024) show that ensembling diverse LLMs produces forecasts statistically indistinguishable from human forecasters, and Tian et al. (2023) document that verbalized LLM prob-

abilities are systematically miscalibrated, motivating post-hoc correction. Benchmark design has matured in parallel: ForecastBench (Karger et al., 2024) and FutureX (Zeng et al., 2025) provide continuously updated evaluations with strict temporal controls, and Paleka et al. (2025) identify systematic pitfalls including temporal leakage and piggybacking on existing forecasts. Despite this progress, existing pipelines treat forecasting as a single-agent process and do not study evidence distribution across a deliberating panel.

Multi-Agent Reasoning and Deliberation. Multi-agent debate has emerged as a general strategy for improving LLM reasoning (Du et al., 2024). Wang et al. (2022) show that sampling diverse chain-of-thought paths and selecting by majority vote improves accuracy, while Wang et al. (2024) propose Mixture-of-Agents, a layered architecture where later agents condition on all prior outputs. Chen et al. (2024) demonstrate that a round-table conference of diverse LLMs with confidence-weighted voting outperforms single-agent baselines across reasoning benchmarks. However, recent theory suggests multi-agent debate is not uniformly beneficial. Choi et al. (2026b) prove that under a Dirichlet-Categorical belief model, standard multi-agent debate forms a *martingale*: expected correctness does not improve over rounds when agents receive identical inputs. Shin (2026) formalize this via the Data Processing Inequality: closed-system deliberation forms a Markov chain, so mutual information with the ground truth can only decrease. Kim et al. (2025) document that LLM errors are 60%+ correlated, implying that naive ensembling hits a non-zero error floor. Our work addresses these limitations by varying the *information structure* of debate rather than its protocol: designed information asymmetry breaks error correlation and converts the martingale into productive belief revision.

Information Aggregation and Collective Intelligence. The conditions under which groups outperform individuals are well studied. Surowiecki (2004) identifies independence, diversity, and decentralized knowledge as key conditions for collective intelligence, while Hong and Page (2004) show that diverse groups can outperform groups of only high-ability individuals. The DeGroot model (DeGroot, 1974) formalizes iterative belief aggregation, and the Delphi method (Dalkey,

1969) applies structured multi-round elicitation to reduce expert disagreement. For probabilistic forecasting, the linear opinion pool (Stone, 1961) and logit-space extremization (Satopää et al., 2014) provide principled aggregation; shrinkage toward a prior is the classical remedy when individual forecasts are overconfident. Meanwhile, social influence can suppress diversity without improving accuracy (Lorenz et al., 2011), and informational cascades may cause agents to ignore private signals once consensus emerges (Bikhchandani et al., 1992). Our framework operationalizes the positive conditions for collective intelligence while explicitly mitigating herding and information loss through evidence partitioning, rationale-based communication, and calibrated aggregation.

3 Methodology

3.1 Problem Formulation

Let q denote a binary forecasting question with resolution date τ and ground truth $y \in \{0, 1\}$. Each question is accompanied by a retrieved evidence corpus $\mathcal{E} = \{e_1, \dots, e_n\}$, where e_i consists of a title, URL, and textual snippet retrieved prior to τ . Given the question q and evidence corpus $\mathcal{E}$, the goal is to estimate $p(y = 1 \mid q, \mathcal{E})$, where $p \in [0, 1]$ denotes the predicted probability that the event resolves positively.

3.2 Theoretical Motivation

We begin by formalizing why information asymmetry improves collective forecasting, drawing on the bias-variance-covariance decomposition of the ensemble mean squared error.

Homogeneous input induces correlated errors. Consider J agents each receiving the same evidence $\mathcal{E}$. Let p_j denote agent j ’s forecast and y the ground truth. The ensemble forecast $\bar{p} = \frac{1}{J} \sum_j p_j$ has mean squared error:

$$\text{MSE}(\bar{p}) = \text{Bias}^2 + \bar{\sigma}^2 - \text{Div}, \quad (1)$$

where $\bar{\sigma}^2 = \frac{1}{J} \sum_j \text{Var}(p_j)$ is the average agent variance and $\text{Div} = \frac{1}{J} \sum_j \text{Var}(p_j - \bar{p})$ is the diversity term (Wood et al., 2023). When all agents observe identical input, their prediction errors $\epsilon_j = p_j - y$ are driven by the same evidence and thus highly correlated: $\text{Cov}(\epsilon_i, \epsilon_j) \approx \text{Var}(\epsilon)$ for $i \neq j$. In this regime the diversity term collapses toward zero, and the ensemble provides no benefit over

a single agent. Recent empirical work confirms this: LLM errors are correlated across providers, and optimal aggregation under full correlation has a non-zero error floor (Kim et al., 2025; Turkmen et al., 2026).

Information asymmetry reduces error correlation. We now show that a public/private evidence partition directly reduces this correlation.

Proposition 3.1 (Diversity-Induced Decorrelation). *Let each agent’s prediction error decompose as $\epsilon_j = \rho \cdot \epsilon^{\text{pub}} + (1 - \rho) \cdot \epsilon_j^{\text{priv}}$, where ϵ^{pub} is the error component driven by shared public evidence, ϵ_j^{priv} is the component driven by agent j ’s private evidence, and $\rho \in [0, 1]$ is the public ratio. Let $\sigma_{\text{pub}}^2 = \text{Var}(\epsilon^{\text{pub}})$ and assume:*

- (i) $\text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}) = 0$ for $i \neq j$ (private-private independence);
- (ii) $\text{Cov}(\epsilon^{\text{pub}}, \epsilon_k^{\text{priv}}) = 0$ for all k (public-private independence);
- (iii) $\text{Var}(\epsilon_j^{\text{priv}}) = \sigma_{\text{priv}}^2$ for all j (symmetric private noise).

Then:

$$\text{Corr}(\epsilon_i, \epsilon_j) = \frac{\rho^2 \sigma_{\text{pub}}^2}{\rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2}. \quad (2)$$

Proof. By bilinearity of covariance and assumptions (i)–(ii), all cross-terms vanish, giving $\text{Cov}(\epsilon_i, \epsilon_j) = \rho^2 \sigma_{\text{pub}}^2$. By assumptions (ii)–(iii), the cross-term in the variance expansion also vanishes, so $\text{Var}(\epsilon_j) = \rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2$, which equals $\text{Var}(\epsilon_i)$ by symmetry. Dividing covariance by $\sqrt{\text{Var}(\epsilon_i)\text{Var}(\epsilon_j)} = \text{Var}(\epsilon_j)$ gives (2). $\square$

Proposition 3.1 shows that inter-agent error correlation decreases monotonically as ρ decreases from 1 to 0. As a concrete instance, at the default $\rho = 0.5$ with equal-variance components ($\sigma_{\text{pub}} = \sigma_{\text{priv}}$), correlation equals $0.25/(0.25 + 0.25) = 0.5$, half the all-public baseline ($\rho=1$, correlation = 1). At $\rho = 0$ (all-private), agents are fully decorrelated; at $\rho = 1$ (all-public, standard debate), correlation reaches its maximum.

Corollary 3.1.1 (Optimal Public Ratio Trade-off). *Decreasing ρ reduces error correlation (improving ensemble gain) but also reduces the shared context necessary for agents to interpret each other’s rationales. An optimal $\rho^* \in (0, 1)$ therefore exists that balances diversity against communicability.*

Rationale sharing enables cross-agent information transfer. Even with disjoint evidence, agents cannot benefit from each other’s private signals unless they communicate their *reasoning*, not just their conclusions. By the Data Processing Inequality (Shin, 2026), a deliberation system where agents only exchange numeric estimates forms a Markov chain $\mathcal{E}_j \rightarrow p_j^{(1)} \rightarrow p_j^{(2)} \rightarrow \dots$, so $I(y; p_j^{(r+1)}) \leq I(y; p_j^{(r)})$: information about the ground truth can only decrease. Sharing rationales breaks this Markov structure: agent i now observes a summary of agent j ’s reasoning over $\mathcal{E}_j^{\text{priv}}$, effectively receiving a lossy projection of j ’s private evidence. This converts the system from closed to open, allowing $I(y; p_j^{(r+1)}) > I(y; p_j^{(r)})$, which is a necessary condition for iterative deliberation to improve over a single round.

3.3 The InfoDelphi Framework

Overview. Figure 2 illustrates the end-to-end InfoDelphi pipeline. Our theoretical analysis identifies three conditions for effective collective forecasting: agents must hold *diverse signals* (to reduce error correlation), share *common ground* (to enable meaningful communication), and exchange *reasoning* (to transfer private information). InfoDelphi instantiates these conditions through three corresponding stages: (1) *evidence routing* partitions $\mathcal{E}$ into a shared public pool and disjoint private subsets via BM25 relevance ranking; (2) *iterative deliberation* enables agents to exchange rationale excerpts across rounds, converting private signals into shared knowledge; (3) *calibrated aggregation* averages final-round forecasts and applies an asymmetric shrinkage that corrects systematic overconfidence.

Evidence Routing. We consider a panel of J agents $\{A_1, \dots, A_J\}$ and a public ratio $\rho \in [0, 1]$. The corpus is partitioned into a shared public subset $\mathcal{E}^{\text{pub}}$ and J disjoint private subsets:

$$\mathcal{E} = \mathcal{E}^{\text{pub}} \cup \bigsqcup_{j=1}^J \mathcal{E}_j^{\text{priv}}, |\mathcal{E}^{\text{pub}}| = \lfloor \rho \cdot |\mathcal{E}| \rfloor. \quad (3)$$

Agent A_j observes the private view $\mathcal{E}_j = \mathcal{E}^{\text{pub}} \cup \mathcal{E}_j^{\text{priv}}$. To maximize information diversity across private subsets, we allocate documents via *BM25 relevance routing*: each document e_i receives a relevance score $s_i = \text{BM25}(e_i, q)$. The top- $|\mathcal{E}^{\text{pub}}|$ scoring documents form $\mathcal{E}^{\text{pub}}$: since these are the

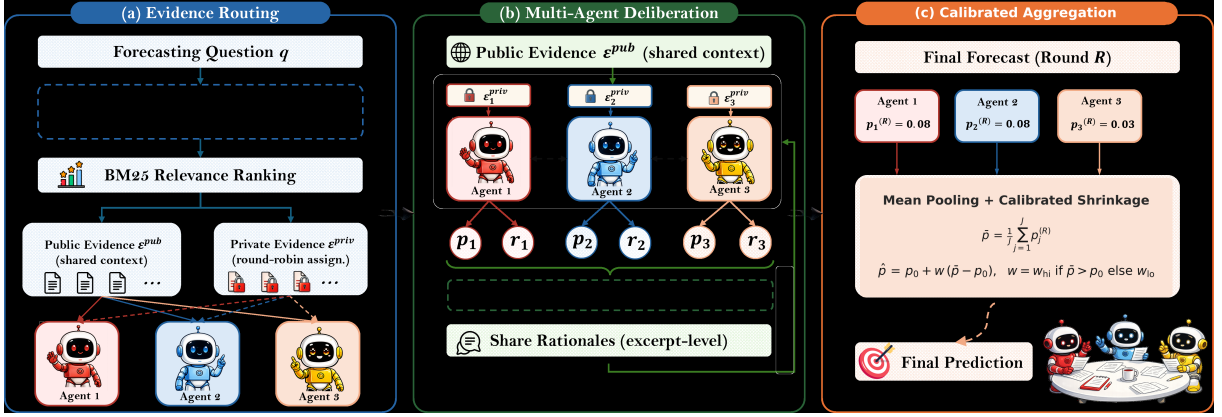


Figure 2: Overview of the **InfoDelphi** framework. (a) **Evidence Routing**: documents are ranked by BM25 relevance and distributed round-robin, constructing a shared public pool and disjoint private subsets for each agent. (b) **Multi-Agent Deliberation**: agents exchange structured forecasts and rationales over two rounds, enabling private information to propagate. (c) **Calibrated Aggregation**: final-round forecasts are averaged and passed through an asymmetric shrinkage transform toward a NO-leaning prior p_0 , correcting systematic overconfidence.

most directly relevant to the question, they serve as the shared “language” through which agents can interpret each other’s reasoning during deliberation. The remaining documents are sorted by s_i and assigned to agents in round-robin order, so that adjacent relevance levels are spread across distinct private views.

Iterative Deliberation with Rationale Sharing. At each round $r \in \{1, \dots, R\}$, agent A_j produces a structured forecast

$$f_j^{(r)} = (p_j^{(r)}, \ell_j^{(r)}, r_j^{(r)}), \quad (4)$$

where $p_j^{(r)} \in [0, 1]$ is a probability estimate, $\ell_j^{(r)} \in \{0, 1\}$ is a categorical prediction, and $r_j^{(r)}$ is a free-text rationale. The agent conditions on its private view and a history summary of prior-round forecasts:

$$f_j^{(r)} \sim \text{LLM}(q, \mathcal{E}_j, h^{(r-1)}), \quad (5)$$

where $h^{(0)} = \emptyset$ and, for $r \geq 2$, $h^{(r-1)} = \{(p_k^{(r-1)}, \ell_k^{(r-1)}, r_k^{(r-1)})\}_{k \neq j}$. Crucially, $h^{(r-1)}$ includes rationale excerpts from other agents, not only their numeric estimates; agents may also quote the specific evidence supporting an argument (evidence pooling), so that a private document cited by one agent becomes visible context for the others. The forecasting prompt instructs agents to reason about base rates and resolution criteria before committing to a probability (the *calibrated prompt*; Appendix H). We study two deliberation protocols: *independent forecasting*, where agents reason solely from their private

view (one round), and *iterative deliberation* (the full InfoDelphi protocol), where agents additionally observe all other agents’ Round 1 forecasts and rationales before producing their final prediction.

Calibrated Aggregation. Given the final-round forecasts $\{p_j^{(R)}\}_{j=1}^J$, InfoDelphi first pools them by the mean, $\bar{p} = \frac{1}{J} \sum_j p_j^{(R)}$, and then applies an *asymmetric shrinkage* toward a prior p_0 :

$$\hat{p} = p_0 + w(\bar{p} - p_0), \quad w = \begin{cases} w_{\text{hi}} & \bar{p} > p_0, \\ w_{\text{lo}} & \bar{p} \leq p_0, \end{cases} \quad (6)$$

with $p_0 = 0.30$, $w_{\text{lo}} = 0.8$, and $w_{\text{hi}} = 0.5$. The design responds to a systematic bias we observe across prompts and datasets: LLM forecasters are overconfident specifically on the *affirmative* side of low-base-rate questions, so forecasts above the prior are shrunk harder ($w_{\text{hi}} < w_{\text{lo}}$) than forecasts below it. The three parameters are fit on one dataset and validated on the other (Section 6.2); the same setting transfers between Polymarket and FutureX without re-tuning. Section 6.2 contrasts this aggregation-layer correction with prompt-level attempts at instilling skepticism, which depress the entire forecast distribution rather than the overconfident region.

4 Experimental Setup

Datasets. Current LLM forecasting systems (Halawi et al., 2024; Turtel et al., 2025) primarily emphasize *retrieval quality*, while

an equally important capability is *information utilization*: how agents interpret, distribute, and collaboratively reason over retrieved evidence. To study this dimension in isolation, we construct **POLYGYM**, a controlled forecasting benchmark of 250 resolved binary Polymarket questions (resolution dates 2025-09 to 2025-12; empirical YES rate 0.232) spanning Sports, Business, Politics, and other domains. Each question is paired with up to 20 pre-retrieved documents collected before market resolution under a strict temporal cutoff, and all methods operate over the same fixed evidence pool, eliminating variability from changing web environments and unstable search results (details in Appendix B). To test generalization beyond Polymarket, we additionally evaluate on 231 resolved questions from the FutureX benchmark (Zeng et al., 2025) (YES rate 0.355), constructed with the same fixed-evidence protocol.

Baselines. We compare against two categories of baselines, all run with the same backbone model and the same evidence pools (details in Appendix E). *Single-agent methods*: Chain-of-Thought (Wei et al., 2022), Self-Consistency (Wang et al., 2022), Sequential Bayesian (Shi et al., 2023), Superforecaster (Karger et al., 2024), and the structured scratchpad pipeline of Halawi et al. (Halawi et al., 2024). *Multi-agent methods with homogeneous input*: Standard Debate (Du et al., 2024), MoA (Wang et al., 2024), Crowd Ensemble (Schoenegger et al., 2024), and AIA Forecaster (Alur et al., 2025). Notably, our Standard Debate implementation shares InfoDelphi’s prompt, deliberation protocol, and mean aggregation, differing only in information structure ($\rho=1$ with full evidence sharing); it therefore doubles as a controlled ablation of information asymmetry.

Metrics and significance. We report the *Brier score* $BS = \frac{1}{N} \sum_{k=1}^N (\hat{p}_k - y_k)^2$ (lower is better) and *accuracy*, the fraction of questions for which $1[\hat{p} \geq 0.5]$ matches y . For every pairwise comparison we run a paired bootstrap over questions (5,000 resamples, fixed seed) and report the 95% confidence interval of the mean Brier difference; a gap is called significant when the interval excludes zero.

Implementation. All experiments use gpt-5.4-mini via the OpenRouter API² as the primary model.³ We use $J = 3$ agents, $R = 2$ deliberation rounds, $\rho = 0.5$, temperature 0.7, and the shrinkage parameters of Eq. (6) for all main results.

5 Main Results

Table 1 reports Brier score and accuracy for all methods on both datasets. Three observations stand out.

InfoDelphi is best on both datasets, and the gaps are significant. The full system attains Brier 0.1575 on POLYGYM-250 and 0.2006 on FutureX-231, the best result in every column. Under the paired bootstrap, the Brier improvement over each of the nine baselines is significant on both datasets (all 95% CIs exclude zero; full intervals in Appendix Table 6). Relative to the strongest baseline on each dataset (Standard Debate), the full system reduces Brier by 13.2% on POLYGYM and 17.6% on FutureX. Even without the shrinkage layer, InfoDelphi’s raw ensemble (Brier 0.1647 / 0.2212) already outperforms every baseline on both datasets, so the ranking does not depend on the calibration layer alone; the ablation in Section 6 quantifies each component’s contribution, and Appendix F reports a control in which the same shrinkage is applied to every baseline.

Homogeneous-input deliberation often underperforms single agents. On FutureX, every multi-agent baseline except Standard Debate is *worse* than the strongest single-agent prompting methods: MoA reaches 0.2944 and Crowd Ensemble 0.2888, compared to 0.2596 for the single-agent Superforecaster prompt. The same pattern appears in weaker form on POLYGYM. This is consistent with the martingale view of homogeneous-input debate (Choi et al., 2026b): when agents share identical evidence, adding agents and rounds mostly redistributes the same information. Information asymmetry is what converts the additional inference cost into additional skill.

²openrouter.ai

³A run of the same InfoDelphi configuration (without the shrinkage layer) with gpt-4o-mini on FutureX-231 reaches Brier 0.237 vs. 0.221 for gpt-5.4-mini, suggesting the architecture is not tied to one backbone; we leave a full cross-model baseline suite to future work.

Table 1: Main results on POLYGYM-250 and FutureX-231. All methods use gpt-5.4-mini over the same fixed pre-resolution evidence pools. †: InfoDelphi (full) is significantly better under a paired bootstrap over questions (5,000 resamples; 95% CI of the Brier difference excludes 0; Appendix Table 6). Best in **bold**.

Method	POLYGYM-250		FutureX-231	
	Brier ↓	Acc ↑	Brier ↓	Acc ↑
<i>Single-agent</i>				
Chain-of-Thought (Wei et al., 2022)†	0.2063	0.6960	0.2611	0.6061
Self-Consistency (Wang et al., 2022)†	0.1965	0.7120	0.2713	0.5498
Sequential Bayesian (Shi et al., 2023)†	0.2523	0.5400	0.3043	0.4848
Superforecaster (Karger et al., 2024)†	0.1937	0.7000	0.2596	0.6364
Halawi et al. (Halawi et al., 2024)†	0.1944	0.7160	0.2654	0.5671
<i>Multi-agent, homogeneous input</i>				
Standard Debate (Du et al., 2024)†	0.1815	0.7200	0.2433	0.6017
MoA (Wang et al., 2024)†	0.1905	0.7200	0.2944	0.5455
Crowd Ensemble (Schoenegger et al., 2024)†	0.1992	0.7000	0.2888	0.5238
AIA Forecaster (Alur et al., 2025)†	0.1897	0.7040	0.2764	0.5498
<i>Ours</i>				
InfoDelphi w/o calibrated shrink	0.1647	0.748	0.2212	0.688
InfoDelphi (full)	0.1575	0.744	0.2006	0.701

Table 2: Component ablation (Brier ↓). Each row removes one component from the full system. “— asymmetry” gives every agent the full evidence pool ($\rho=1$, full sharing) with the same calibrated prompt, i.e., the Standard Debate configuration of our pipeline. Paired 95% CIs for each removal are given in the text.

Configuration	POLYGYM	FutureX
InfoDelphi (full)	0.1575	0.2006
— calibrated shrink (mean agg.)	0.1647	0.2212
— deliberation (round 1 only)	0.1698	—
— information asymmetry	0.1815	0.2433

Calibration matters as much as protocol. Sequential Bayesian, the weakest baseline on both datasets (0.2523 / 0.3043), fails primarily through calibration: its sequential updates drift toward extreme probabilities. Conversely, InfoDelphi’s shrinkage layer converts a 0.1647 raw score into 0.1575 on POLYGYM (-0.0072 , paired CI $[-0.0180, +0.0029]$) and 0.2212 into 0.2006 on FutureX (-0.0206 , CI $[-0.0417, -0.0013]$), with the same parameters on both datasets. Section 6.2 shows that this correction cannot be reproduced at the prompt level.

6 Analysis

6.1 Why Information Asymmetry Helps

Component ablation. Table 2 removes one component at a time. Replacing calibrated aggregation with a plain mean costs 0.007 / 0.021 Brier on the two datasets. Removing deliberation (using round-1 forecasts only) costs 0.005 on POLYGYM

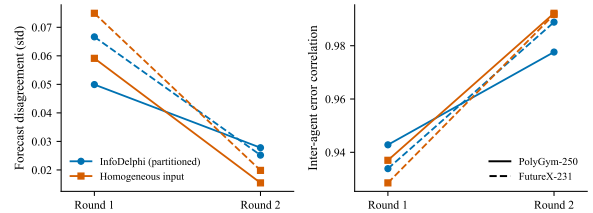


Figure 3: Deliberation dynamics with and without information asymmetry. Left: mean per-question standard deviation of agent forecasts; right: mean pairwise inter-agent error correlation. Deliberation under homogeneous input collapses forecast diversity (herding); partitioned evidence preserves independent signal after deliberation on both datasets.

(paired $\Delta = -0.0052$, CI $[-0.0184, +0.0083]$): the second round helps, but the effect is small relative to question-level noise and not individually significant. The largest and most robust factor is information structure: giving every agent the full evidence pool with full sharing (the Standard Debate configuration, which shares our prompt and aggregation) degrades the raw ensemble from 0.1647 to 0.1815 on POLYGYM (paired $\Delta = -0.0169$, CI $[-0.0314, -0.0029]$) and from 0.2212 to 0.2433 on FutureX ($\Delta = -0.0221$, CI $[-0.0515, +0.0081]$), significant on POLYGYM and directionally consistent on FutureX.

Herding dynamics. Figure 3 traces *how* homogeneous input wastes deliberation. In round 1, per-question forecast disagreement is comparable for the two information structures (sampling noise

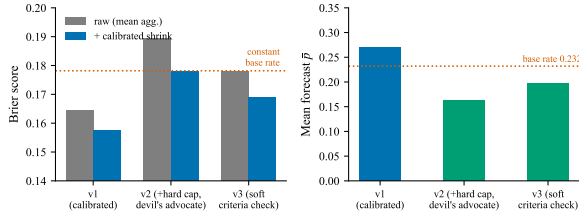


Figure 4: Prompt-level vs. aggregation-level calibration on POLYGYM-250. Left: Brier score of three calibration prompts, with and without aggregation-level shrinkage; the dotted line is a constant base-rate predictor. Right: mean forecast $\bar{p}$ against the empirical base rate. Skeptical prompt wording (v2, v3) depresses the entire forecast distribution below the base rate, whereas asymmetric shrinkage corrects overconfidence where it occurs.

dominates). After one round of deliberation, however, disagreement under homogeneous input collapses by 74–79% (e.g., 0.059 $\rightarrow$ 0.015 on POLYGYM), and mean pairwise error correlation rises to 0.99: the panel has effectively merged into a single agent. With partitioned evidence, agents converge far less (0.050 $\rightarrow$ 0.028) and retain a lower error correlation (0.978), preserving the diversity term of Eq. (1) that ensemble averaging needs. This is direct empirical evidence for the mechanism behind Proposition 3.1: what matters is not deliberation itself but whether deliberation has diverse inputs to propagate.

6.2 Calibration: Aggregation Layer vs. Prompt

Our agents, like LLM forecasters generally (Tian et al., 2023), are overconfident on the affirmative side: high-probability YES forecasts hit far less often than stated. We compared two remedies. The *prompt-level* route rewrites the forecasting prompt to demand skepticism: v2 adds a devil’s-advocate agent and a hard confidence cap; v3 replaces the cap with a soft resolution-criteria checklist. The *aggregation-level* route keeps the original calibrated prompt (v1) and applies the asymmetric shrinkage of Eq. (6).

Figure 4 shows the outcome. Both prompt interventions *hurt*: v2 degrades raw Brier from 0.1647 to 0.1896 — worse than predicting the constant base rate (0.1782) — and v3 lands at 0.1780. The right panel shows why: prompt-level skepticism cannot be targeted at the overconfident band, so it depresses the whole distribution, pushing the mean forecast from 0.270 down to 0.163 (v2) and 0.198 (v3) against a base rate of 0.232, sacrific-

Table 3: Targeted retrieval augmentation on the 106 Sports questions of POLYGYM-250 (InfoDelphi full; paired bootstrap vs. original evidence, 5,000 resamples). Neither augmentation yields a significant improvement.

Evidence	Brier ↓	AUC ↑	Δ Brier [95% CI]
Original	0.1908	0.696	—
+ opponent identity	0.1890	0.690	-0.0018 [-0.010, +0.006]
+ opponent odds/form	0.1873	0.720	-0.0035 [-0.013, +0.005]

ing true-YES questions in the mid-band. Shrinkage applied on top of each prompt improves all three variants, but the best combination by a clear margin is the unmodified prompt plus shrinkage (0.1575). The parameters fit on Polymarket transfer to FutureX unchanged (0.2212 $\rightarrow$ 0.2006, accuracy 0.688 $\rightarrow$ 0.701), while re-fitting on FutureX and applying to Polymarket recovers the same setting, indicating the correction captures a model-level bias rather than a dataset artifact. We conclude that overconfidence corrections belong in the aggregation layer, where they can act asymmetrically on the miscalibrated region, not in the prompt.

6.3 What Deliberation Cannot Fix

Per-domain analysis shows Sports (106 of 250 POLYGYM questions) is the weakest domain. A natural hypothesis is missing information: match-level evidence often lacks the opponent’s identity or recent form. We tested this with two targeted augmentations: decoding the opponent from the market URL slug (adding it as metadata), and retrieving pre-cutoff news about the decoded opponent’s odds and form under a three-layer leakage control. Both augmentations were consumed — 52–53% of agent forecasts cite the new documents, and 43% of retrieved documents contain numeric odds — yet neither yields a measurable gain (Table 3): paired differences are -0.002 and -0.004 with CIs straddling zero, and untreated control subsets moved as much as treated ones. We interpret sports outcomes as evidence-limited at the *model* level: the residual uncertainty is aleatoric given the kind of evidence a text corpus provides, and no amount of deliberation or retrieval within this class of evidence should be expected to remove it. This delimits the scope of our contribution: information asymmetry improves the *utilization* of informative evidence; it cannot create information that the corpus does not

contain.

7 Conclusion

We presented **InfoDelphi**, a multi-agent forecasting framework built on a simple but overlooked insight: collective reasoning only outperforms individual reasoning when participants hold genuinely different information. Treating information asymmetry as a first-class design principle — and correcting overconfidence in the aggregation layer rather than the prompt — yields the best Brier score among ten systems on two fixed-evidence benchmarks, with paired-bootstrap significance against every baseline. The accompanying analysis localizes the mechanism (herding collapse under homogeneous input), the calibration lesson (asymmetric shrinkage transfers across datasets; prompt-level skepticism backfires), and the boundary of the approach (evidence-limited domains do not respond to better utilization). These findings suggest that in multi-agent LLM systems, *what each agent reads* deserves the same design attention as *how agents talk*.

Limitations

Theoretical assumptions. Proposition 3.1 rests on three idealizing assumptions: a linear error decomposition, public–private independence, and symmetric private noise. In practice, documents retrieved for the same question share topical correlations, partially violating independence and making the decorrelation bound approximate; LLMs, as non-linear functions of their inputs, need not satisfy the fixed-weight decomposition. Consistently, the empirical error correlations in Figure 3 remain high (~ 0.94) even with partitioned evidence, because model-level correlation persists regardless of evidence allocation.

Effect sizes and scope. The deliberation-round effect is small relative to question-level noise and not individually significant, and the advantage over the strongest homogeneous-input baseline, while significant on POLYGYM, is directional on FutureX; when the shrinkage transform is granted to all baselines (Appendix F), InfoDelphi remains best on both datasets but several pairwise gaps are no longer individually significant. Our primary experiments use a single backbone model (gpt-5.4-mini), binary questions, and fixed evidence pools; live-retrieval settings, open-ended questions, and heterogeneous model pan-

els remain untested. The shrinkage parameters, while validated across our two datasets, are not guaranteed to transfer to models or question distributions with different bias profiles. API cost scales linearly with agents $\times$ rounds (1,500 forecasting calls for a full POLYGYM-250 run at $J=3$, $R=2$), which limited exploration of larger panels and longer deliberation.

Ethics Considerations

This work presents a forecasting framework and does not directly enable disinformation or harm beyond risks inherent to LLM-based systems generally. However, automated probability estimates generated at scale could in principle be misused to manipulate market sentiment or create false impressions of consensus. We recommend that InfoDelphi be used as a decision-support tool with human oversight in high-stakes deployment contexts.

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A Discussion

Key insight. Our central finding is that *designed information asymmetry* is necessary for effective multi-agent deliberation. When all agents receive identical evidence, iterative discussion provides little measurable benefit, as agents merely reinforce correlated beliefs rather than contribute complementary knowledge (Figure 3). This aligns with the martingale characterization of homogeneous-input debate and suggests that the widely adopted practice of giving all agents the same context fundamentally limits the value of multi-agent interaction.

Future work. Natural extensions include (1) adaptive routing that dynamically adjusts evidence allocation based on emerging disagreement, (2) more expressive routing algorithms beyond BM25, such as dense retrieval or learned semantic partitioning, (3) learned aggregation replacing the fixed shrinkage parameters, (4) evaluation on open-ended and multi-choice forecasting tasks, and (5) heterogeneous model panels that leverage model-level diversity in addition to evidence-level asymmetry.

Potential risks. Designed information asymmetry encourages agents to maintain distinct beliefs, but it may also amplify biased or misleading evidence partitions. If private evidence subsets contain low-quality, adversarial, or outdated documents, agents may become confidently polarized around incorrect conclusions rather than self-correcting through deliberation. In addition, rationale sharing introduces a potential attack surface, where persuasive but flawed rationales propagate errors across agents. Future deployments should incorporate evidence verification, source reliability estimation, and safeguards against adversarial persuasion.

B POLYGYM Benchmark Details

Motivation. Agentic forecasting systems for prediction markets require two distinct capabilities: (1) *retrieval*, identifying relevant information from the web, and (2) *utilization*, reasoning over that information to produce calibrated probability estimates. Existing benchmarks predominantly evaluate these capabilities jointly (Table 4), making it difficult to determine whether performance gains arise from better retrieval or better collective reasoning. Live benchmarks ([Karger](#)

et al., 2024; Zeng et al., 2025) additionally introduce variability from changing web environments and unstable evidence availability, and retrieval-augmented systems continue to exhibit degradation over time, suggesting that access to information alone does not solve the utilization challenge (Dai et al., 2025). POLYGYM therefore fixes the evidence pool: all methods observe the same pre-retrieved corpus, while different agents may observe different subsets of it. Conceptually, this parallels controlled retrieval settings in QA research (Chen et al., 2021; Yang et al., 2018) that separate retrieval quality from downstream reasoning.

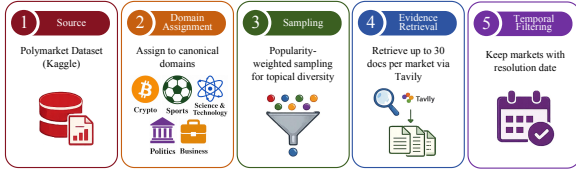


Figure 5: POLYGYM data construction pipeline.

Construction. POLYGYM is derived from a large-scale Polymarket dataset (Semedov, 2025) consisting of approximately 40,000 events and 100,000 prediction markets. Events are assigned to canonical domains using metadata fields, and popularity-weighted sampling ensures topical diversity. After temporal filtering (below), the benchmark contains 250 resolved binary markets: Sports (106), Business (28), Politics (23), Entertainment (6), Geopolitics (5), and other domains (82). The empirical YES rate is 0.232. All questions and evidence documents are in English.

Evidence. For each market, we retrieve up to 20 documents via the Tavily search API⁴ using queries derived from the question text (mean 19.5 documents per question). A strict temporal cutoff ensures all evidence precedes the market resolution date. Retrieved documents are deduplicated and truncated to 500 characters.

Temporal filtering. To reduce contamination from LLM prior knowledge, we retain only markets with resolution dates between 2025-09-09 and 2025-12-02, after the primary model’s knowledge cutoff.

Licenses and intended use. The source Polymarket dataset (Semedov, 2025) is licensed under

⁴<https://www.tavily.com/>

the Open Data Commons Database Contents License (DbCL) v1.0. Evidence documents retrieved via the Tavily Search API are used solely for non-commercial research purposes under its API terms of service and are not independently redistributed. POLYGYM is intended for research use only. The accompanying code is released under the MIT License.

C Proof of Proposition 3.1

We provide the complete derivation. Let $\epsilon_j = \rho \cdot \epsilon^{\text{pub}} + (1 - \rho) \cdot \epsilon_j^{\text{priv}}$ for agent $j \in \{1, \dots, J\}$.

Covariance between agents. For $i \neq j$, expanding by bilinearity of covariance:

$$\begin{aligned} \text{Cov}(\epsilon_i, \epsilon_j) &= \rho^2 \text{Var}(\epsilon^{\text{pub}}) \\ &\quad + \rho(1-\rho) \text{Cov}(\epsilon^{\text{pub}}, \epsilon_j^{\text{priv}}) \\ &\quad + \rho(1-\rho) \text{Cov}(\epsilon_i^{\text{priv}}, \epsilon^{\text{pub}}) \\ &\quad + (1-\rho)^2 \text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}). \end{aligned} \quad (7)$$

By the independence assumptions, $\text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}) = 0$ and $\text{Cov}(\epsilon^{\text{pub}}, \epsilon_j^{\text{priv}}) = 0$, so $\text{Cov}(\epsilon_i, \epsilon_j) = \rho^2 \text{Var}(\epsilon^{\text{pub}})$.

Individual variance. Since ϵ^{pub} and ϵ_j^{priv} are independent, the cross term vanishes: $\text{Var}(\epsilon_j) = \rho^2 \text{Var}(\epsilon^{\text{pub}}) + (1-\rho)^2 \text{Var}(\epsilon_j^{\text{priv}})$.

Correlation. Let $\sigma_{\text{pub}}^2 = \text{Var}(\epsilon^{\text{pub}})$ and assume agents are symmetric ($\text{Var}(\epsilon_j^{\text{priv}}) = \sigma_{\text{priv}}^2$ for all j), which implies $\text{Var}(\epsilon_i) = \text{Var}(\epsilon_j)$ and simplifies the correlation to

$$\text{Corr}(\epsilon_i, \epsilon_j) = \frac{\rho^2 \sigma_{\text{pub}}^2}{\rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2}. \quad (8)$$

In the homogeneous baseline where $\rho = 1$: $\text{Corr}(\epsilon_i, \epsilon_j) = 1$. As ρ decreases, correlation decreases monotonically, reaching 0 when $\rho = 0$. At our default $\rho = 0.5$ with equal-variance components, $\text{Corr} = 0.5$, a 50% reduction relative to $\rho = 1$. $\square$

D Proof of Corollary 3.1.1

Define the *communicability* between two agents as the mutual information between their forecast errors, $C(\rho) := I(\epsilon_i; \epsilon_j)$. When $\rho = 0$, the private components are independent by assumption, so $C(0) = 0$. For $\rho > 0$, agents share the common component ϵ^{pub} , so $C(\rho) > 0$. By the Data

Setup	Static	Reproducible	Time Control	Fixed Evidence	Realistic	Agent Study
ForecastBench (Karger et al., 2024)	×	×	✓	×	✓	×
FutureX (Zeng et al., 2025)	×	×	✓	×	✓	×
PolyBench (Cheng et al., 2026)	×	×	✓	×	✓	×
Prediction Arena (Zhang et al., 2026)	×	×	✓	×	✓	✓
Zou et al. (Zou et al., 2022)	✓	×	×	×	×	×
Halawi et al. (Halawi et al., 2024)	✓	×	×	×	×	×
POLYGYM (Ours)	✓	✓	✓	✓	✓	✓

Table 4: Comparison of forecasting evaluation setups. POLYGYM provides fixed evidence and full reproducibility, enabling controlled study of information utilization and multi-agent deliberation.

Processing Inequality (Shin, 2026), $C(\rho)$ upper-bounds the information any function of ϵ_i (including agent i ’s rationale) can provide to agent j : at $\rho = 0$, deliberation cannot transfer information between agents.

Ensemble MSE decomposition. Our objective is to find $\rho^* = \arg \min_{\rho \in [0,1]} L(\rho)$. The J -agent ensemble MSE decomposes as

$$L(\rho) = \underbrace{\frac{\sigma^2(\rho)}{J} [1 + (J-1)r(\rho)]}_{\text{variance term } V(\rho)} + \underbrace{D(\rho)^2}_{\text{bias term}}, \quad (9)$$

where $\sigma^2(\rho) = \rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2$ is the individual agent variance and $r(\rho)$ is the inter-agent correlation from Proposition 3.1. We model $D(\rho)$ as a continuous, non-increasing function of $C(\rho)$, with $D(0) = D_0$ (single-agent bias, since deliberation cannot transfer information when $C(0) = 0$) and $D(\rho) < D_0$ for all $\rho > 0$.

Boundary analysis. At $\rho = 1$: $r(1) = 1$, so $V(1) = \sigma_{\text{pub}}^2$, equal to single-agent variance with no ensemble gain. For any $\rho = 1 - \varepsilon$ with $\varepsilon > 0$: $r(1-\varepsilon) < 1$, so $V(1-\varepsilon) < V(1)$, while $D(1-\varepsilon) \approx D(1)$ by continuity. Therefore $L(1-\varepsilon) < L(1)$, and $\rho = 1$ is not a minimizer.

At $\rho = 0$: $\sigma^2(0) = \sigma_{\text{priv}}^2$, $r(0) = 0$, and $C(0) = 0$. Differentiating, $\left. \frac{d\sigma^2}{d\rho} \right|_{\rho=0} = -2\sigma_{\text{priv}}^2 < 0$ and $\left. \frac{dr}{d\rho} \right|_{\rho=0} = 0$ (the numerator of r vanishes to second order at $\rho = 0$), so

$$\left. \frac{dV}{d\rho} \right|_{\rho=0} = -\frac{2\sigma_{\text{priv}}^2}{J} < 0. \quad (10)$$

By continuity there exists $\varepsilon_V > 0$ such that $V(\varepsilon) < V(0)$ for $\varepsilon \in (0, \varepsilon_V)$; simultaneously, $C(\varepsilon) > 0$ implies $D(\varepsilon) < D_0$. Combining, $L(\varepsilon) < L(0)$ for sufficiently small $\varepsilon > 0$, so $\rho = 0$ is not a minimizer.

Existence of interior optimum. Since neither boundary is a minimizer and $L(\rho)$ is continuous on the compact interval $[0, 1]$, the minimum is attained at some $\rho^* \in (0, 1)$. $\square$

E Experimental Details

E.1 Baseline Descriptions

Single-Agent Methods.

- **Chain-of-Thought** (Wei et al., 2022): one agent receives all evidence and produces a forecast with chain-of-thought reasoning.
- **Self-Consistency** (Wang et al., 2022): the same CoT prompt is sampled 5 times at temperature 0.7 and predictions are averaged.
- **Sequential Bayesian** (Shi et al., 2023): evidence items are processed one at a time by $K=5$ sequential agents, each updating the previous agent’s probability estimate.
- **Superforecaster** (Karger et al., 2024): a structured reasoning prompt that instructs the model to decompose the question, estimate base rates, update with evidence, and perform a calibration check.
- **Halawi et al.** (Halawi et al., 2024): a 7-step structured scratchpad prompt (rephrase, pro/con arguments, aggregate, calibration check) run 3 times, combined via a trimmed mean that downweights the prediction furthest from the median by 50%.

Multi-Agent Methods (Homogeneous Input).

- **Standard Debate** (Du et al., 2024): 3 agents all receive the full evidence corpus, deliberate over 2 rounds sharing predictions and rationales, with final predictions aggregated by mean. As noted in Section 4, our implementation shares InfoDelphi’s prompt and protocol and differs only in information structure.
- **MoA** (Wang et al., 2024): 3 proposer agents independently forecast with full evidence; a 4th aggregator agent receives all proposals

with rationales and produces a final synthesized prediction.

- **Crowd Ensemble** (Schoenegger et al., 2024): 5 independent agents each receive all evidence and predict; the median prediction is the final forecast.
- **AIA Forecaster** (Alur et al., 2025): 10 independent agents forecast with full evidence, and a supervisor agent identifies disagreements and produces a final prediction with a confidence label; on low confidence the system falls back to the mean of the 10 agents.

E.2 Computational Cost

A full InfoDelphi run on POLYGYM-250 issues $250 \times 3 \times 2 = 1,500$ forecasting calls plus lightweight aggregation (no additional model calls); baselines range from 250 calls (CoT) to 2,750 (AIA). Cost scales linearly with agents $\times$ rounds.

F Shrinkage Control Experiment

Because the calibrated-shrink transform of Eq. (6) is a post-hoc map on the final probability, it can be granted to any method. Table 5 applies the identical transform (same parameters, no per-method tuning) to every baseline. Three facts emerge. (1) Shrinkage helps every method on both datasets, confirming that the affirmative-side overconfidence it corrects is a model-level bias rather than an artifact of our pipeline. (2) The ranking is preserved: InfoDelphi remains the best system on both datasets. (3) The comparison is honest about effect sizes: with all methods shrunk, several pairwise gaps — notably against Superforecaster, MoA, and Standard Debate on POLYGYM — are no longer individually significant at $n = 250$, although the gaps against 4 of 9 baselines on POLYGYM and 6 of 9 on FutureX remain so. We therefore attribute InfoDelphi’s advantage to the combination of information structure and calibrated aggregation, and report both raw and shrunk baselines so the reader can weigh each factor.

G Significance Tests

Table 6 reports the paired bootstrap intervals behind the daggers in Table 1: for each baseline, the mean question-level Brier difference between InfoDelphi (full) and the baseline, with the 95% percentile interval over 5,000 resamples (seed 0).

H Prompt Templates

All agents receive the same system prompt and structured user prompt; the only difference across agents is their assigned evidence subset.

System prompt (calibrated, v1 — used in all main results).

```
You are a forecaster for binary
(Yes/No) prediction markets.
Use only the evidence provided.
Prefer high-credibility sources
when they conflict.
Follow this procedure:
1. BASE RATE: before weighing
the evidence, estimate what
fraction of questions like this
one resolve YES. "Will X happen
by <date>" questions usually
require a specific chain of
events to complete in time,
so they resolve NO more often
than the surrounding discussion
suggests.
2. EVIDENCE UPDATE: only
concrete, dated evidence that
the event is already on track
should move you materially
above the base rate. Topical
coverage, speculation, or
enthusiasm is not evidence for
YES.
3. SHRINK: if the evidence is
indirect, incomplete, or stale,
move your estimate toward the
base rate -- not toward 0.5.
4. CALIBRATION CHECK: out of
100 questions where you would
give this exact probability,
how many should actually resolve
YES? Adjust if your number feels
like a hedge or a headline.
You must respond with valid JSON
only, no other text before or
after.
```

Prompt variants of the calibration study (Section 6.2). Variant v2 inserts a *resolution-criteria* step and a *high-confidence gate*: before giving $p_{\text{yes}} \geq 0.7$, the agent must verify every resolution criterion against cited, dated evidence and otherwise “stay at or below 0.6”; it additionally runs a devil’s-advocate agent. Variant v3 removes the hard cap and instead appends a soft re-check to the calibration step: when any resolution criterion is supported only by analogy, “move back toward the base rate.”

User prompt (round 1).

```
Question: {question}
Evidence:
[1] doc_id=doc_001 | source=...
| title=... | content: ...
```

Table 5: Control experiment: applying our calibrated-shrink transform ($p_0=0.30$, $w_{lo}=0.8$, $w_{hi}=0.5$) to every baseline’s output. Shrinkage improves all methods; the ranking is preserved and InfoDelphi remains best on both datasets, although several pairwise gaps are no longer individually significant (n.s.). Δ is the paired Brier difference of InfoDelphi (full) vs. the shrunk baseline (negative favors ours).

Method	POLYGYM-250			FutureX-231		
	raw	+shrink	Δ [95% CI]	raw	+shrink	Δ [95% CI]
Chain-of-Thought	0.2063	0.1738	−0.016 [−0.029, −0.003]	0.2611	0.2158	−0.015 [−0.033, +0.003] n.s.
Self-Consistency	0.1965	0.1706	−0.013 [−0.025, −0.002]	0.2713	0.2218	−0.021 [−0.038, −0.004]
Sequential Bayesian	0.2523	0.2025	−0.045 [−0.059, −0.030]	0.3043	0.2411	−0.041 [−0.060, −0.021]
Superforecaster	0.1937	0.1665	−0.009 [−0.021, +0.004] n.s.	0.2596	0.2146	−0.014 [−0.031, +0.003] n.s.
Halawi et al.	0.1944	0.1693	−0.012 [−0.024, +0.000] n.s.	0.2654	0.2224	−0.022 [−0.039, −0.004]
Standard Debate	0.1815	0.1650	−0.008 [−0.015, +0.000] n.s.	0.2433	0.2064	−0.006 [−0.022, +0.011] n.s.
MoA	0.1905	0.1670	−0.010 [−0.021, +0.002] n.s.	0.2944	0.2326	−0.032 [−0.049, −0.015]
Crowd Ensemble	0.1992	0.1701	−0.013 [−0.025, −0.000]	0.2888	0.2309	−0.030 [−0.048, −0.012]
AIA Forecaster	0.1897	0.1651	−0.008 [−0.019, +0.004] n.s.	0.2764	0.2223	−0.022 [−0.039, −0.003]
InfoDelphi (full)	0.1647	0.1575	—	0.2212	0.2006	—

Table 6: Paired bootstrap comparison of InfoDelphi (full) against each baseline (5,000 resamples over questions, seed 0). Δ Brier < 0 favors InfoDelphi; all 95% CIs exclude zero.

Baseline	POLYGYM-250 Δ Brier [95% CI]	FutureX-231 Δ Brier [95% CI]
Chain-of-Thought	−0.0488 [−0.0756, −0.0217]	−0.0606 [−0.0958, −0.0258]
Self-Consistency	−0.0390 [−0.0619, −0.0149]	−0.0707 [−0.1032, −0.0381]
Sequential Bayesian	−0.0948 [−0.1209, −0.0681]	−0.1037 [−0.1402, −0.0684]
Superforecaster	−0.0362 [−0.0620, −0.0097]	−0.0591 [−0.0941, −0.0248]
Halawi et al.	−0.0370 [−0.0615, −0.0123]	−0.0648 [−0.0966, −0.0338]
Standard Debate	−0.0240 [−0.0416, −0.0064]	−0.0427 [−0.0761, −0.0114]
MoA	−0.0331 [−0.0573, −0.0086]	−0.0938 [−0.1295, −0.0580]
Crowd Ensemble	−0.0418 [−0.0662, −0.0167]	−0.0882 [−0.1224, −0.0534]
AIA Forecaster	−0.0323 [−0.0566, −0.0077]	−0.0758 [−0.1103, −0.0409]

```
[2] doc_id=doc_002 | source=...
| title=... | content: ...
...
Respond with exactly one JSON
object, no other text. Schema:
{"p_yes": <0-1>, "label":
"YES"|"NO", "rationale": "...",
"evidence_used": ["doc_id",
...]}
```

User prompt (round 2). The round-2 prompt appends a history summary of all other agents’ round-1 outputs after the evidence:

```
Previous round summary (other
agents):
Agent 1: p_yes=0.850 (YES) --
The evidence suggests...
Agent 2: p_yes=0.120 (NO) --
Based on the pricing data...
Round mean p_yes=0.533.
Consider whether you agree or
disagree with the above.
```

With evidence pooling enabled, rationales cite documents by `doc_id` together with the concrete fact each contributes, so privately-held documents referenced in a rationale become visible context for other agents in the next round.